Introduction to Machine Learning

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Outline

Machine Learning And Data Mining

The nuts and bolts

- Models
- Algorithms
- Theory
- Practice

Workflow

- Scientific question
- Formalisation of the problem
- Data collection
- Analysis and model selection

Types of statistics / machine learning problems

- Classification
- Regression
- Density estimation
- Reinforcement learning

Machine learning

Data Collection

- Downloading a clean dataset from a repository
- Scraping data from the web
- Conducting a survey
- Performing experiments, and obtaining measurements.

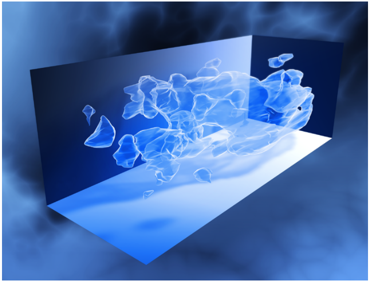
Modelling

- Simple: the bias of a coin
- Complex: a language model.
- The model depends on the data and the problem

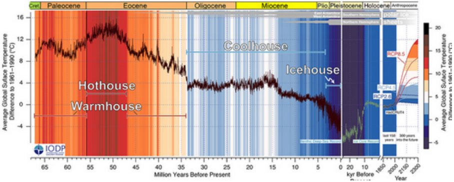
Algorithms and Decision Making

- We want to use models to make decisions.
- Decisions are made every step of the way.
- Both humans and algorithms can make decisions.

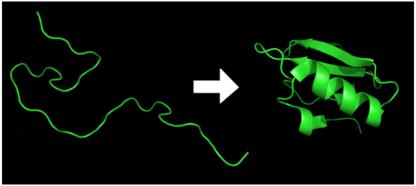
The main problems in machine learning and statistics



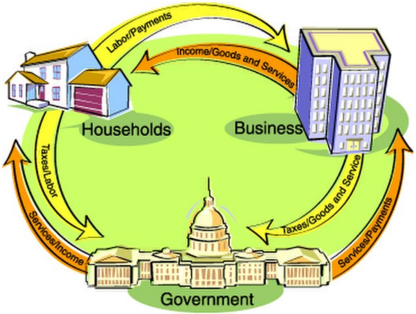
Dark Matter



Climate Modelling

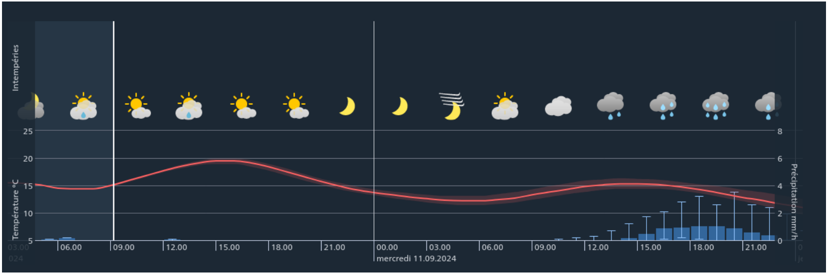


Protein Folding



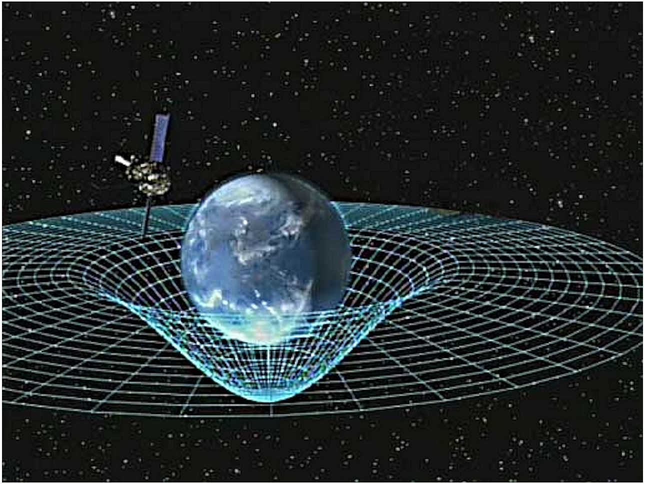
Economic Policy

Prediction



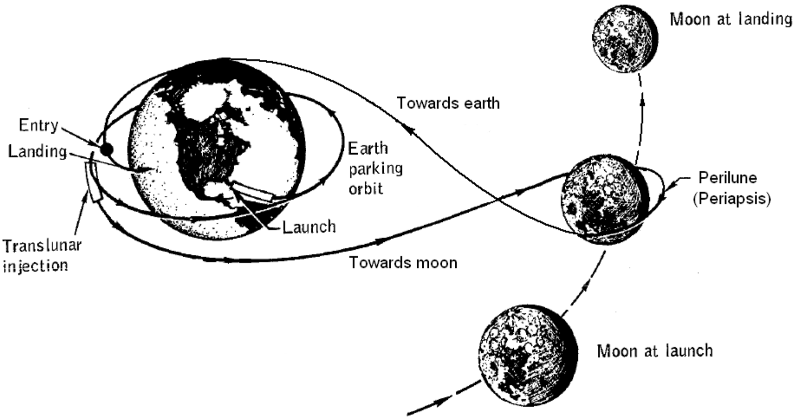
- Will it rain tomorrow?
- How much will bitcoin be worth next year?
- When is the next solar eclipse?

Inference



- What is the law of gravitation?
- Where is the spaceship now?
- Does my poker opponent have two aces?

Decision Making



- What data should I collect?
- Which model should I use?
- Should I fold, call, or raise in my poker game?
- How can I get a spaceship to the moon and back?

./fig/artemis.gif

The need to learn from data

Problem definition

- What problem do we need to solve?
- How can we formalise it?
- What properties of the problem can we learn from data?

Data collection

- Why do we need data?
- What data do we need?
- How much data do we want?
- How will we collect the data?

Modelling and decision making

- How will we compute something useful?
- How can we use the model to make decisions?

Course Material

Moodle

- Assignments and proejct
- Additional reading material
- Asking questions

Course Github

- .org files for notes, PDF for slides
- source code for examples

Course literature

- An Introduction to Statistical Learning with Python
- Book chapters will be mentioned in the course

Assignments, teaching and questions

Course structure

- In-class exercises.
- Take-home assignments (ungraded).
- Quizzes.
- In-class group work.

Other questions

- Use Moodle for technical/administrative questions: That way everybody gets the same information.
- Use email for personal problems or extra help, if the moodle is not enough.
- Complicated questions can be answered at the next lecture.

Office hours

- Tuesdays 13:15-14:00: book with an email to avoid clashes.
- Email me for an appointment outside those hours.

The project pipeline

- Builds on in-class exercises and take-home assignment
- Every group must create a github repository and commit regularly.
- A short presentation will be given.

Key points

- Define a scientific question.
- Identify relevant variables and their relationships.
- Generate data from counterfactual simulations.
- Analyse the data from the different counterfactuals.

Problem definition

- Example: Health, weight and height

Example (Health questions regarding height and weight)

- What is a normal height and weight?
- How are they related to health?
- What variables affect height and weight?

Define a research question

Find a non-sensitive variable that we can easily measure via a survey, e.g. related to sleep, smoking, exercise, food, politics, sports, hobbies etc.

- Discuss in small groups and post suggestions
- We then vote for what to measure

Data collection

Think about which variables we need to collect to answer our research question.

Necessary variables

The variables we need to know about

- Weight
- Height
- Dependent: (health/vote/opinion/salary)

Auxiliary variables

Measurable factors related to the variables of interest

Possible confounders

+Hidden factors that might affect variables. We will formalise these dependencies with graphical models later in the course — for now, just try to list them.

Class data and variables

- The class enters their data into the excel file.

- Pay attention to the variables we wish to measure

Privacy

- Is the use of a pseudonym sufficient to hide your identity?

Variables

The class data looks like this

First Name	Gender	Height	Weight	Age	Nationality	Smoking
Lee	M	170	80	20	Chinese	10
Fatemeh	F	150	65	25	Turkey	0
Ali	Male	174	82	19	Turkish	0
Joan	N	5'11	180	21	American	4

- $\boldsymbol{X}$: Everybody's data
- x_t: The t-th person's data
- $x_{t,k}$: The k-th feature of the t-th person.
- $\boldsymbol{x}_k$: Everybody's k-th feature
- Anything strange with this data?

Raw versus neat data

- Neat data: $x_t \in \mathbb{R}^n$
- Raw data: web pages, handwritten text, graphs, data packets, missing/incorrect values, etc

Types of learning problems

Unsupervised learning (unconditional estimation)

- Predict the gender of an unknown individual.
- Predict the height.
- Predict the height and weight?

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- Classification: Can we predict gender from height/weight?
- Regression: Can we predict weight from height and gender?
- In both cases we predict output variables from input variables

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Variables

- Input variables: aka features, predictors, independent variables
- Output variables: aka response, dependent variables, labels, or targets.
- The input/output dichotomy only exists in some prediction problems.

Looking at the class data

Reading the class data

```
import pandas as pd
X = pd.read_excel("data/class.xlsx")
X["Height"]       # bx_k : the Height feature, for everybody
X.loc[2]          # x_t  : individual 2, all features
```

- A DataFrame is not the same as a numpy array.
- Columns keep their names ("Height")
- Rows can be accessed through index.

Summarising the class data

```
X.hist()
import matplotlib.pyplot as plt
plt.show()
```

- Check the histograms for inconsistnet raw values: does anything look wrong?

Pandas and DataFrames

- Data in pandas is stored in a DataFrame
- DataFrame is not the same as a numpy array.

Core libraries

```
import pandas as pd
import numpy as np
```

Series: A sequence of values

```
# From numpy array:
s = pd.Series(np.random.randn(3),  index=["a", "b", "c"])
# From dict:
d = {"a": 1, "b": 0, "c": 2}
s = pd.Series(d)
# accessing elemets
s.iloc[2] #element 2
s.iloc[1:2] #elements 1,2
s.array # gets the array object
s.to_numpy() # gets the underlying numpy array
```

DataFrames

Constructing from a numpy array

```
data = np.random.uniform(size = [3,2])
df = pd.DataFrame(data, index=["John", "Ali", "Sumi"],
         columns=["X1", "X2"])
```

Constructing from a dictionary

```
d = {  "one": pd.Series([1, 2], index=["a", "b"]),
       "two": pd.Series([1, 2, 3], index=["a", "b", "c"])}
df = pd.DataFrame(d)
```

Access

```
X["First Name"] # get a column
X.loc[2] # get a row
X.at[2, "First Name"] # row 2, column 'first name'
X.loc[2].at["First Name"] # row 2, element 'first name' of the serie
X.iat[2,0] # row 2, column 0
```

Modelling single variables

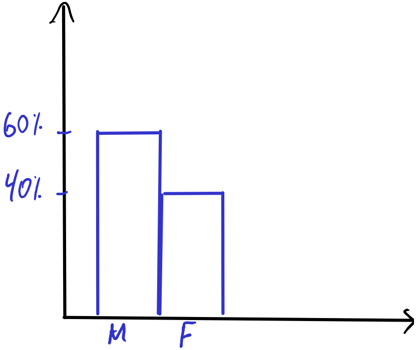


Figure: $x \in \mathbb{N}$

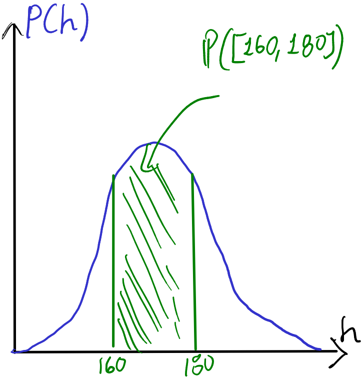


Figure: $x \in \mathbb{R}$

Means using python

Example (Calculating the mean of our class data)

```
X.mean()              # mean of every column of the class DataFrame
X["Height"].mean()    # just the height column
np.mean(X["Weight"]) # equivalent, via numpy
```

- The mean here is fixed because we calculate it on the same data.
- If we were to collect new data then the answer would be different.

Example (Calculating the mean of a random variable)

```
import numpy as np
y = np.random.gamma(170, 1, size=20)  # 20 fresh samples -- not the
y.mean()
np.mean(y)
```

- The mean is random, so we get a different answer every time we re-run this.

Probabilities: Basic notions.

Probability space

- Defined on subsets of Ω, a random outcome space.
- Subsets A of Ω, $A \subseteq \Omega$, are called events.

Probability axioms

A probability P, is a function from subsets[1] of Ω to $[0, 1]$ and it functions like an area:

- $P(\Omega) = 1$ (whatever happens must be in Ω).
- $P(A) \geq 0$ is the probability of $A \subseteq \Omega$.
- If A, B are mutually exclusive ($A \cap B = \emptyset$), then:

$$P(A \cup B) = P(A) + P(B).$$

[1] Formally, an algebra

Meaning of proability

Probabilities are usually subjective (exception: quantum physics).

Subjective probability rule

If we believe an event A is more likely than B then $P(A) > P(B)$.

Properties that follow.

- Implication: If A always happens if B happens, then every rational person should believe A is at least as likely as B.
- Transitive property: If A is more likely than B and B is more likely than C then A is more likely than C.

Special cases

The outcomes or universe Ω depends on what we are interested in, and how we generate the random outcomes.

Discrete Ω

- Consider throwing two coins. Then possible outcomes are $\Omega = \{HH, HT, TH, TT\}$ for heads or tails for each coin.
- Consider throwing one die. Then possible outcomes are $\Omega = [6] = \{1, 2, 3, 4, 5, 6\}$.
- Consider selecting an individual in Switzerland by randomly generating an AVS number: $\Omega = [10^{12}]$.

Continous Ω

Consider throwing a die inside a 1×1 box. Then possible outcomes are in $\Omega = [0, 1]^2 \times [0, 360]^3$, because we need:

- 2 $[0, 1]$ variables for the location
- 3 $[0, 360]$ variables for the angle of rotation

Which is the most natural outcome space for throwing a die?

Random variables

Even though the most natural outcome space for a die throw is $[6]$, we actually observe the die as a 3D object in $[0,1]^2 \times [0,360]^3$. So we map the outcome in Ω onto $[6]$ if we want to play dice, i.e. we construct a random variable (RV) $x : [0,1]^2 \times [0,360]^3 \to [6]$.

Dice game

- Here consider $\Omega = [6]$ and $P(\omega) = 1/6$ for every outcome $\omega \in \Omega$.
- Define the RV $x : \Omega \to \{0, 1\}$ as

$$x(\omega) = \begin{cases} 1, & \text{if } \omega >= 5 \\ 0, & \text{if } \omega < 5 \end{cases}$$

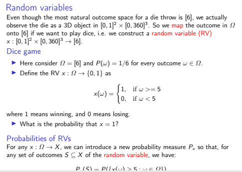
where 1 means winning, and 0 means losing.

- What is the probability that $x = 1$?

Probabilities of RVs

For any $x : \Omega \to X$, we can introduce a new probability measure P_x so that, for any set of outcomes $S \subseteq X$ of the random variable, we have:

$$P_x(S) = P(\{x(\omega) \geq 5 : \omega \in \Omega\})$$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Alternative definition

- An equivalent definition: $x(\omega) = \mathbb{I}\{\omega \geq 5\}$,

uses the indicator function $\mathbb{I}\{\}$ which takes the value 1 when its argument is true and 0 otherwise.Probability of $x = 1$ Since we are only interested in $x = 1$, then $S = \{1\}$. As a shorthand, we usually write $P_x(1)$ instead of $P_x(\{1\})$.

$$P_x(1) = P(\{x(\omega) = 1 : \omega \in \Omega\})$$

Here, $\{x(\omega) = 1 : \omega \in \Omega\}$ is the set of values $\omega \in \Omega$ for which $x(\omega) = 1$. These are $\{5, 6\}$ in our example. Consequently,

$$P_x(1) = P(\{5, 6\}) = P(5) + P(6) = 1/6 + 1/6 = 1/3.$$

This second equality follows from the third axiom of probability, because $\{5\}$ and $\{6\}$ are mutually exclusive.

One variable: expectations

Definition (The expected value)

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Definition (The sample mean)

- i.i.d. variables $x_1, \ldots, x_t, \ldots, x_T$: with $x_t = x(\omega_t)$, $\omega_t \sim P$.

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Definition (The sample mean)

- i.i.d. variables $x_1, \ldots, x_t, \ldots, x_T$: with $x_t = x(\omega_t)$, $\omega_t \sim P$.
- The sample mean of $x_1, \ldots, x_T$ is

$$\frac{1}{T} \sum_{t=1}^{T} x_t$$

The sample mean is $O(1/\sqrt{T})$-close to $\mathbb{E}_P[x_t]$ with high probability.

Reminder: expectations of random variables

A gambling game

- [a] With probability 1%, you win 100 CHF
- [b] With probability 40%, you win 20 CHF.
- [c] Otherwise, you win nothing

What are the expected winnings if you play this game?

Reminder: expectations of random variables

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Solution

- Let x be the amount won, then $x(a) = 100, x(b) = 20, x(c) = 0$.
- We need to calculate

$$\mathbb{E}_P(x) = \sum_{\omega \in \{a,b,c\}} x(\omega)P(\omega) = x(a)P(a) + x(b)P(b) + x(c)P(c)$$

- $P(c) = 59\%$, as $P(\Omega) = 1$. Substituting,

$$\mathbb{E}_P(x) = 1 + 8 + 0 = 9.$$

Models

Models as summaries

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Example models

- A numerical mean
- A linear classifier
- A linear regressor
- A deep neural network
- A Gaussian process
- A large language model

Estimates and decisions

We always need to make decisions based on some estimates.

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Estimate the bias of a coin

- I give you a coin that, lands with some fixed probability on heads.
- You are allowed to experiment with the coin.
- I will pay you 1 CHF if you guess the throw correctly
- Otherwise you pay me x CHF.
- How much should I ask you to pay for the bet to be fair?
- What do you need to know to determine this?

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Example (If the coin is fair)

- If the coin is fair, then you only have 50% proability of guessing correctly.
- If you bet x CHF, your expected return is x

The Bernoulli distribution

The Bernoulli distribution

Definition (Bernoulli distribution)

We say that $x \in \{0, 1\}$ has Bernoulli distribution with parameter θ and write

$$x \sim \mathrm{Bernoulli}(\theta),$$

when

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Example (Applications of the Bernoulli distribution)

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- Classification errors.

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Example (Applications of the Bernoulli distribution)

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Exercise: The expected value

If x is Bernoulli with parameter θ, then what is the expected value of

- The variable $f(x) = x - 1$?
- The variable $g(x) = (x - 1)^2$?

Two-variable models

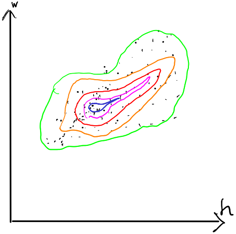


Figure: $x \in \mathbb{R}^2$

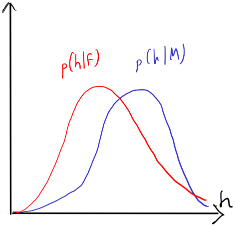
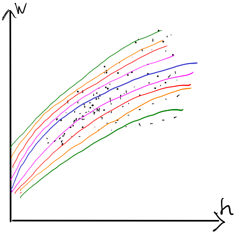
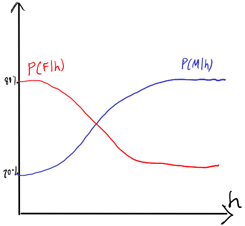


Figure: $x \in \mathbb{N} \rightarrow y \in \mathbb{R}$





Predicting y from x, discrete case.

Consider two variables, x, y. We can either care about

- $\mathbb{E}[y|x]$ the expectation of y for all x.
- $\mathbb{P}[y|x]$ the distribution of y for all x.

Probability table for $P(x, y)$

$P(x, y)$	y = 0	y = 1
x = 0	54%	6%
x = 1	16%	24%

- How can we graph this?
- What is $P(x)$?

Conditional probability table for $P(y|x)$

$P(y \mid x)$	y = 0	y = 1
x = 0	90%	10%
x = 1	40%	60%

- What is $\mathbb{E}[y \mid x]$?

Distributions of two variables

In this setting, both x and y have a Bernoulli distribution. If we use a model whereby x is sampled first, and then y, then we can define two Bernoulli distributions. The first, for x is unconditional, while the second, for y, depends on the value of x:

$$x \sim \mathrm{Bernoulli}(\theta)$$
$$y \mid x \sim \mathrm{Bernoulli}(\phi_x).$$

In our example, $\phi_0 = 0.1$ and $\phi_1 = 0.6$.

Homework

Probability table for $P(x, y)$

$P(x,y)$	y = -1	y = 0	y = 1
x = 0	10%	20%	10%
x = 1	30%	20%	10%

Given the above table, calculate

- $P(x)$
- The conditional probability table for $P(y|x)$.
- $\mathbb{E}_P[f]$ where $f(x, y) = x + y$.
- $\mathbb{E}_P[g]$ where $g(x, y) = xy$.
- $\mathbb{E}[y|x]$ for all values of x.

Two variables: conditional expectation

The height of different genders

The conditional expected height

$$\mathbb{E}[h \mid g = 1] = \sum_{\omega \in \Omega} h(\omega) P[\omega \mid g(\omega) = 1]$$

The empirical conditional expectation

$$\mathbb{E}[h \mid g = 1] \approx \frac{\sum_{t : g(\omega_t) = 1} h(\omega_t)}{|\{t : g(\omega_t) = 1\}|}$$

Python implementation

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Python implementation

```
h[g==1] / sum(g==1)
## alternative
import numpy as np
np.mean(h[g==1])
```

Populations, samples, and distributions

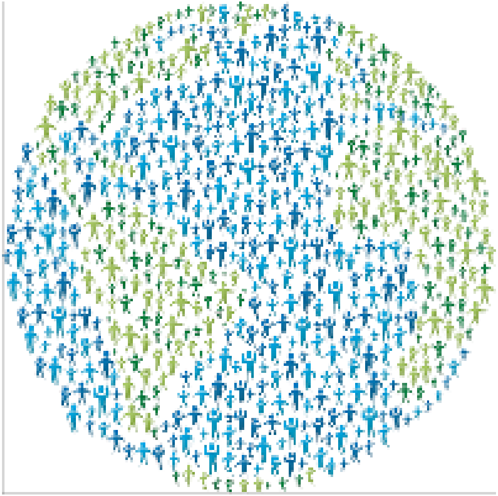

Figure: The world population

Figure: A sample

Statistical assumptions

Independent, Identically Distributed data

- $\omega_t \sim P$: individuals $\omega_t \in \Omega$ are drawn from some distribution P
- $\boldsymbol{x}_t \triangleq \boldsymbol{x}(\omega_t)$ are some features of the t-th individual
- Here we are interested in properties of the unknown distribution P.

Representative sample from a fixed population

- Finite population $\Omega = \{\omega_1, \omega_2, \ldots, \omega_N\}$
- A subset $S \subset \Omega$ of size $T < N$ is selected with a uniform distribution, i.e. so that

$$P(S) = T/N, \qquad \forall S \subset \Omega.$$

- Here we are interested in statistics of the unknown population Ω.
- We assume an underlying distribution P for convenience.
- We can tried both cases essentially the same.

Learning from data

Unsupervised learning

- Given data $x_1, \ldots, x_T$.
- Learn about the data-generating process.
- Example: Estimation, compression, text/image generation

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Online learning

- Sequence prediction: At each step t, predict x_{t+1} from $x_1, \ldots, x_t$.
- Conditional prediction: At each step t, predict y_{t+1} from $x_1, y_1 \ldots, x_t, y_t, {\color{red}x_{t+1}}$

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Reinforcement learning

Learn to act in an unknown world through interaction and rewards

Data, models, and reproducibility.

Training data

- Calculations, optimisation
- Data exploration

Data, models, and reproducibility.

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Validation data

- Fine-tuning
- Model selection

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Simulation

- Interactive performance comparison
- White box testing

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Real-world testing

- Actual performance measurement

Model selection

- Train/Test/Validate
- Cross-validation
- Simulation

Course Contents

Models

- k-Nearest Neighbours.
- Linear models and perceptrons.
- Multi-layer perceptrons (aka deep neural networks).
- Bayesian Networks

Algorithms

- (Stochastic) Gradient Descent.
- Bayesian inference.

Reproducibility

- Modelling assumptions
- Interactions and feedback

Fairness

- Implicit biases in training data
- Fair decision rules and meritocracy

Assignment

- See Moodle

Reading for this week

ISLP Chapter 1